Publication List

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Five Selected Publications with Statements of Significance

[1] Wentao Zhang, Z. Wu, Y. Fan, W. Hua, and M. Cheng, "Analysis and Improvement of DC Winding Current Dynamics in Wound Field Switched Flux Machine," *IEEE Trans. Ind. Appl.*, vol. 60, no. 4, pp. 6023–6032, Jul.–Aug. 2024. (IEEE Xplore)

This paper develops the first analytical framework linking converter PWM switching to field-winding current oscillations in wound-field machines via air-gap field modulation. The work identifies the harmonic mechanism responsible for uncontrolled current dynamics and proposes a suppression strategy that restores stable, controllable winding-current behaviour. The contribution is directly relevant to the control of next-generation magnet-free traction drives, where predictable current dynamics are essential for field-oriented control and sensorless operation. The methodology underpins three related patents on induced-voltage suppression.

[2] Wentao Zhang, Z. Wu and W. Hua, "High-Fidelity Lumped Parameter Thermal Model for Oil-Cooling Hairpin Winding PM Machines in EVs," *IEEE Trans. Transport. Electrific.*, vol. 12, no. 2, pp. 2978–2991, Apr. 2026. (IEEE Xplore)

This paper presents the first high-fidelity lumped-parameter thermal network with fluid-solid coupled oil-cooling for hairpin-winding PM traction machines, validated against AVL dynamometer measurements across the full speed-torque operating map. The model provides the thermal plant representation essential for model-based thermal management and derating control in high-speed EV drive systems. The work is part of a series building on the prize-winning *n*-layer flat-wire winding design paper [J5] (**2025 IEEE Trans. Transportation Electrification Prize Paper Award, 2nd Prize**), extending the programme from electromagnetic-mechanical design to coupled thermal characterisation of the Inovance traction-machine prototypes.

[3] Wentao Zhang, Z. Wu, Y. Fan, W. Hua, and M. Cheng, "A Stable and Computationally Efficient Spatial Harmonic Model for Predicting the DC Winding Induced Voltage in WFSF Machine," *IEEE Trans. Ind. Appl.*, vol. 59, no. 4, pp. 3966–3977, Jul.–Aug. 2023. (IEEE Xplore)

This paper develops a computationally efficient spatial-harmonic inductance model that captures the full harmonic spectrum of wound-field switched-flux machines. The model resolves numerical instabilities present in prior subdomain approaches while maintaining sufficient accuracy for drive-system simulation. Its computational efficiency makes it suitable for real-time implementation in model-based controllers and digital-twin applications, where conventional finite-element analysis is too slow. The work provides the dynamic machine model needed for model-based control of wound-field drives.

[4] Wentao Zhang, Zhongze Wu and Wei Hua, "A Deep Learning Based Magnetic Field Surrogate Modeling Method for Interior Permanent Magnet Machine in EVs," in Proc. Int. Conf. Elect. Mach. (ICEM), Funchal, Madeira, Portugal, 2026. (Accepted) (view attached paper)

This paper proposes an implicit physics-knowledge embedding approach in which a deep neural network, augmented with a Fourier coordinate-encoding layer, maps machine design parameters to the full magnetic vector potential distribution of a dual-V interior PM machine. Unlike conventional surrogates that predict scalar performance only, the method recovers the spatial field, from which torque, flux linkage and iron loss are obtained by standard post-processing. The model achieves $R^2 = 0.9996$ for torque and 0.9730 for iron loss on unseen designs, with inference time reduced by more than 94% against finite-element analysis, opening a path toward rapid design-space exploration.

[5] Wentao Zhang, Z. Wu, L. Jin, Y. Fan, W. Hua, and M. Cheng, "Analysis and Multi-Objective Optimization of the Hybrid Excitation Switched Flux Machine," *IEEE Trans. Ind. Appl.*, vol. 61, no. 2, pp. 2996–3006, Mar.–Apr. 2025. (IEEE Xplore)

This paper presents a multi-objective optimisation framework for hybrid excitation machines, balancing electromagnetic torque density, efficiency, flux-regulation capability and induced-voltage suppression across

the design space. The optimisation couples analytical and finite-element models within a systematic trade-off methodology. The work demonstrates the application of advanced optimisation to electrified drive-system design and received the **CIEEC Best Paper Award (2023)** in its conference form. The methodology is directly transferable to system-level optimisation of integrated motor-drive systems.

Complete Publication Record

Journal Papers

[J1] **Wentao Zhang**, W. Hua, Z. Wu, G. Zhao, Y. Wang, and W. Xia, "Analysis of DC Winding Induced Voltage in Wound-Field Flux-Switching Machine With Air-Gap Field Modulation Principle," *IEEE Trans. Ind. Electron.*, vol. 69, no. 3, pp. 2300–2311, Mar. 2022.

[J2] **Wentao Zhang**, Z. Wu, W. Hua, Z. Zhu, Z. Wu, J. Hu, et al., "Reduction of Open-Circuit DC Winding Induced Voltage and Torque Pulsation in the Wound Field Switched Flux Machine by Stator Axial Pairing of Tooth Tips," *IEEE Trans. Ind. Appl.*, vol. 58, no. 2, pp. 1976–1990, Mar.–Apr. 2022.

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[J4] **Wentao Zhang**, Z. Wu, Y. Fan, W. Hua, and M. Cheng, "Analysis and Improvement of DC Winding Current Dynamics in Wound Field Switched Flux Machine," *IEEE Trans. Ind. Appl.*, vol. 60, no. 4, pp. 6023–6032, Jul.–Aug. 2024.

[J5] **Wentao Zhang**, Z. Wu and W. Hua, "Design of *n*-Layer Flat Wire Winding in Permanent Magnet Machine for Electric Vehicles," *IEEE Trans. Transport. Electrific.*, vol. 11, no. 3, pp. 7798–7810, 2025.

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[J7] **Wentao Zhang**, Z. Wu, Z. Wang, W. Hua, Y. Fan, Z. Xu, et al., "Design Tradeoff Between Flux Regulation Capability and DC Winding Induced Voltage in Hybrid Excitation Switched Flux Machine," *IEEE Trans. Ind. Appl.*, vol. 61, no. 5, pp. 7198–7208, Sep. 2025.

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[J10] **Wentao Zhang**, Z. Wu and W. Hua, "High-Fidelity Lumped Parameter Thermal Model for Oil-Cooling Hairpin Winding PM Machines in EVs," *IEEE Trans. Transport. Electrific.*, vol. 12, no. 2, pp. 2978–2991, Apr. 2026.

[J11] Z. Wu, **Wentao Zhang**, Y. Fan, W. Hua, and M. Cheng, "A Transient-State Lumped Parameter Thermal Model for Brushless Wound Field Switched Flux Machines," *IEEE Trans. Transport. Electrific.*, vol. 10, no. 1, pp. 224–238, Mar. 2024.

[J12] Z. Wu, Z. Zhu, C. Wang, W. Hua, K. Wang, and **Wentao Zhang**, "Influence of Rotor Iron Bridge Position on DC-Winding-Induced Voltage in Wound Field Switched Flux Machine Having Partitioned Stators," *Chin. J. Electr. Eng.*, vol. 7, no. 3, pp. 20–28, Sep. 2021.

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Conference Papers

[C1] **Wentao Zhang**, Zhongze Wu and Wei Hua, "A Deep Learning Based Magnetic Field Surrogate Modeling Method for Interior Permanent Magnet Machine in EVs," in Proc. Int. Conf. Elect. Mach. (ICEM), Funchal, Madeira, Portugal, 2026. (**Accepted**)

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Patents

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[C2] Zhongze Wu, **Wentao Zhang**, Lai Jin, and Wei Hua, "Disk Brushless Electric Excitation Magnetic Field Modulation Motor With Double-Row Stator Structure," CN118100572A, 2024.

[C3] Zhongze Wu, **Wentao Zhang**, Huilin Huang, and Wei Hua, "Method for Constructing a High-Precision Motor Performance Database Based on a Dual-Model Strategy," CN117972924A, 2024.

[C4] Zhongze Wu, **Wentao Zhang**, Wei Hua, and Zhixiang Zou, "Method for Establishing General Lumped Parameter Thermal Network Model of Motor," CN117094088A, 2023.

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[C6] Zhongze Wu, **Wentao Zhang**, Junxiang Liao, Ying Fan, Wei Hua, and Ming Cheng, "Excitation Winding Induced Pulse Voltage Suppression Method Based on Harmonic Current Injection," CN115208245A, 2022.

[C7] Zhongze Wu, **Wentao Zhang**, Ying Fan, Wei Hua, and Ming Cheng, "High-Precision Field Path Coupling Design Method for Brushless Electric Excitation Magnetic Field Modulation Motor," CN114781227A, 2022.

[C8] Zhongze Wu, **Wentao Zhang**, Wei Hua, and Ziqiang Zhu, "Method for Weakening the Induced Pulsating Voltage of a Brushless Electric Excitation Magnetic Field Modulated Motor," CN114189185A, 2022.

[C9] Wei Hua, **Wentao Zhang**, Zhongze Wu, and Shichuan Ding, "Method for Reducing Torque Ripple and Induced Voltage of Electric Excitation Flux Switching Motor," CN111769697A, 2020.

Appendix: Attached Paper

The following pages contain the full text of the accepted ICEM 2026 paper referenced as selected publication [4]:

Wentao Zhang, Zhongze Wu and Wei Hua, "A Deep Learning Based Magnetic Field Surrogate Modeling Method for Interior Permanent Magnet Machine in EVs," in Proc. Int. Conf. Elect. Mach. (ICEM), Funchal, Madeira, Portugal, 2026. (Accepted).

A Deep Learning Based Magnetic Field Surrogate Modeling Method for Interior Permanent Magnet Machine in EVs

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***Abstract*—In this paper, a deep learning based magnetic field surrogate modeling method is proposed for the interior permanent magnet (IPM) machine. The analysed method utilizes an implicit physics knowledge embedding method to model the magnetic vector potential through the data-driven method. Different from traditional surrogate modeling method, which is based on the mapping between design space to the output performance, the proposed method maps the design space of the IPM machine to the magnetic vector potential, and therefore the torque, flux-linkage and iron loss can be calculated through conventional post-processing method. The analysis shows that the proposed modeling approach could achieve better torque and iron loss prediction under different topology and current advance angle excitation conditions, which features superior performance than the traditional surrogate model. Moreover, the proposed method could also provide high fidelity magnetic field prediction, whilst the inference time is >94% reduced compared with the finite element analysis.**

Keywords—Deep learning, electric vehicle (EV), interior permanent magnet (IPM) machine, surrogate model

I. INTRODUCTION

The electric machine has become an important power conversion device in the transition to the sustainable transportation. The rapid development, analysis and optimization for the EV used traction machine has raised considerable attention in both academia and the industry [1], [2]. Although the Maxwell's equations can accurately describe magnetic field physics in the electric machine, the field distribution characteristic and the complicated material characteristic has increased the analysis difficulty [3].

The finite element (FE) analysis is one of the most accurate electric machine analysis method, which could handle different topologies and non-linear material properties. However, the FE based analysis method still suffer from high calculation time, which is time-consuming in the large scale optimization [4].

The FE analysis alternative methods can be divided into two types, the equivalent magnetic circuit/network method [5], [6] and the analytical methods [7], [8], [9]. The equivalent magnetic circuit/network methods are based on the analogy between the magnetic field and the electrical circuit. The modeling accuracy of these methods are based on the discretization level of the electrical machine topology [5]. However, those methods is difficult to capture the geometry details of the electric machine. The analytical methods includes the conformal mapping [7], [8] and subdomain methods [9], which could achieve rapid prediction of the magnetic field of the electric machine. However, the analytical method is difficult to account the non-linear characteristic and the geometry details.

The recently emerging data-driven methods offer attractive options for the electric machine performance prediction tools. It could accelerate the search for the pareto front during the multi-objective optimization [10]. Response surface [11] and surrogate modeling [12], [13] are two representative cases for the optimization assistance methods. However, those methods are based on the direct mapping between the design space and the output performance. Therefore, the accuracy of the data-driven modeling methods are based on the dataset size and the high quality sampling. The effectiveness of such methods can be challenging under high dimensional optimization scale. In recent years, the artificial intelligence (AI) / machine learning (ML) analysis approach have successfully applied in the fluid dynamics and partial differential equation (PDE) disciplines, which shows superior performance for capturing the science physics [14], [15], [16]. Since the magnetic fields of the electrical machine are governed by the Maxwell's equation, which fundamentally belongs to the PDE family, the AI/ML analysis approach are highly promising alternatives for the FE analysis.

Based on the deep learning (DL) modeling approach, this paper aims at fulfilling this research gap to propose a magnetic field surrogate modeling method for the dual-V layer interior permanent magnet (IPM) machine. The spatial distribution of the magnetic field can be modeled based on the proposed method, and based on which the torque and iron loss can be predicted in high accuracy and generalization capability.

This paper is organized as follows. In Section II, the analysed IPM machine and the detailed analysis setting are given. In Section III, the methodology of the proposed surrogate modeling method are introduced and the performance prediction method are compared with the traditional methods. Moreover, the field fidelity analysis are also compared between proposed and the FE methods in Section III, including both the magnetic field distribution and the iron loss distribution. The conclusion and discussion are given in Section IV.

II. THE ANALYSED IPM MACHINE TOPOLOGIES

A. Topology

The analysed IPM machine has a dual-V shape PM rotor, which is a commonly used PM machine topology. The illustration of the machine topology is shown in Fig. 1. It

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should be noted that the moving band mesh is used in the analysed IPM machine. The finite element analysis model is built based on the in-house code package in Python 3.11.

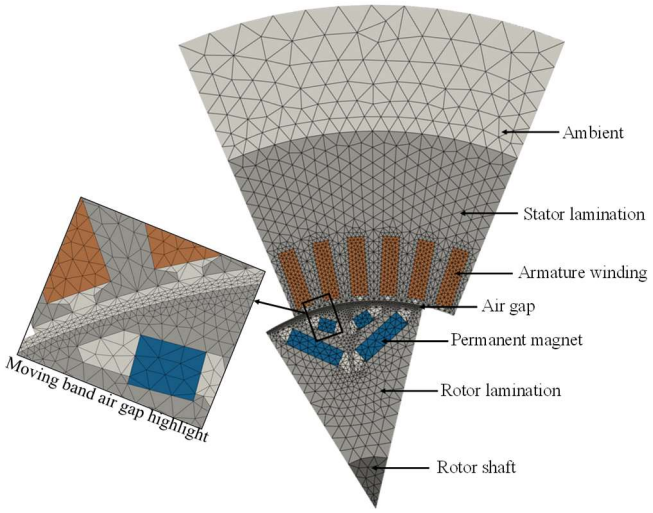


Fig. 1. Analysed fractional model of dual-V shape IPM machine.

B. Surrogate Model Parameters

The surrogate modeling of the analysed IPM machine will include different topology design parameters, rotor position and the x-, y-coordinate positions. It should be noted that the outer diameter and the current density are fixed at 180 mm and 17.68 Arms/mm^2, which is feasible in the modern traction applications. The current phase advance angle is also included in the design parameters, which can be scanned to find the optimum torque under maximum torque per Ampere control mode. Both fixed parameter and the variables in the surrogate model are given in TABLE I.

TABLE I FIXED/VARIABLE PARAMETERS FOR ANALYSED IPM MACHINE

Item	Unit	Value/Bounds
Stator outer diameter	mm	180
Stack length	mm	100
Armature winding current density	Arms/mm^2	17.68
Slot packing factor	-	0.7
Single side airgap width	mm	1
Rotor outer diameter	mm	(60, 120)
Stator slot width	mm	(1.0, 6.0)
Stator slot height	mm	(12.0, 30.0)
Stator slot opening width	mm	(1.0, 2.5)
1st layer PM position coefficient	-	(0.4, 0.6)
1st layer PM array angle coefficient	-	(0.75, 1.0)
1st layer PM array width	mm	(2.5, 5.0)
1st layer PM bottom flux barrier width	mm	(0.5, 3.0)
1st layer PM top flux barrier width	mm	(0.5, 3.0)
2nd layer PM position coefficient	-	(0.4, 0.8)
2nd layer PM array angle coefficient	-	(0.7, 0.9)
2nd layer PM width	mm	(2.5, 5.0)
2nd layer PM bottom flux barrier width	mm	(0.5, 3.0)
2nd layer PM top flux barrier width	mm	(0.5, 3.0)
Current advance angle	deg.	(90, 180)

III. DEEP LEARNING BASED SURROGATE MODELING

A. Dataset Preparation and Modeling Workflow

The deep learning surrogate modeling approach is data-driven based analysis method, and the generation of the dataset should be clarified in this subsection. The illustration of the dataset preparation and the modeling workflow is shown in Fig. 2.

The variational range of the dataset is shown in TABLE I. To ensure uniform distribution in the feasible design region, Latin Hypercube Sampling (LHS) method is applied in this paper. Since the geometry has possibility to fail under specific design parameters, 45000 designs are set for the LHS in this paper, and 4520 designs are calculated successfully.

During the calculation, the scalar value of the average torque, iron loss and torque waveform are calculated and recorded as the performance of the IPM cases. Moreover, the magnetic vector potential distribution under every mesh nodes are also recorded at every rotor position during the calculation.

For the conventional surrogate modeling method, only design parameters and the scalar performance value are used for modeling, which could suffer from overfitting and poor generalization ability.

As for the deep learning based surrogate modeling approach, design parameters and the magnetic vector potential distribution are applied for the training of the network. It should be noted that the scalar performance are unknown for the deep learning based surrogate modeling method in this paper. The proposed method will use the conventional post-processing method as that in the finite element analysis, and the accuracy is dependent on the prediction of the magnetic vector potential. The illustration of the magnetic vector potential and governing equation can be found in Appendix.

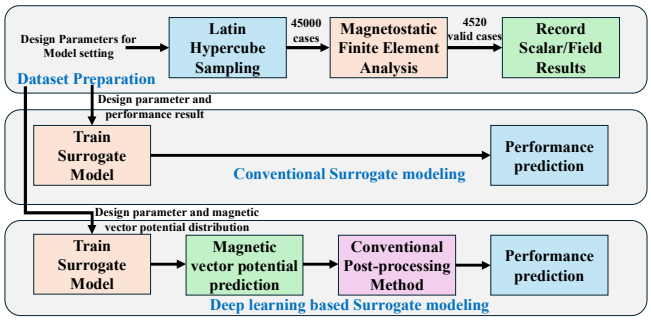
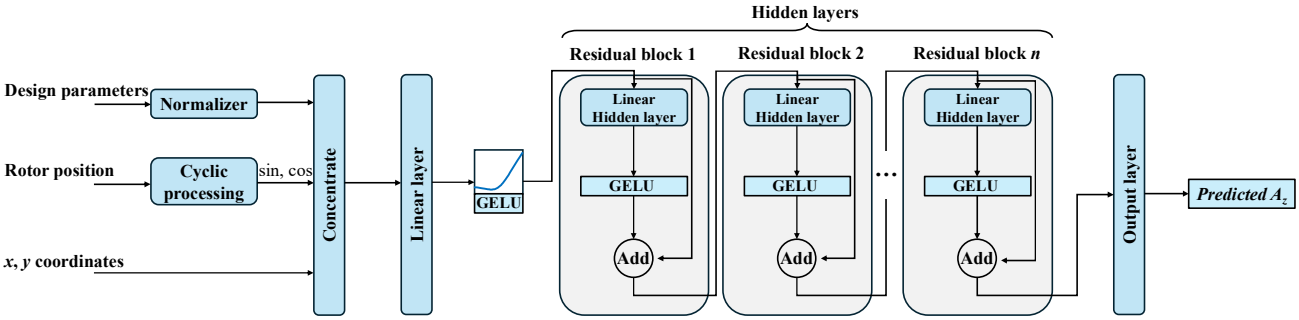


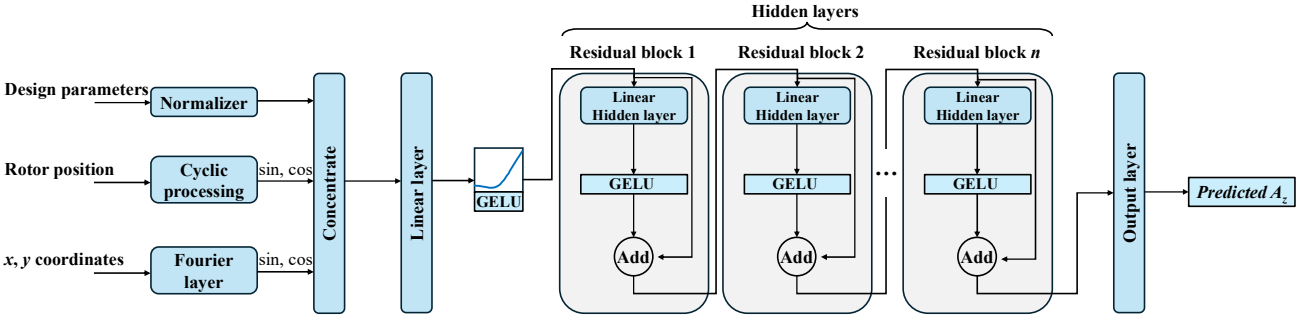
Fig. 2. Illustration of the modeling workflow.

B. Deep Learning Model Structure and Training

The proposed deep learning based surrogate model has three types of input, the design parameters as depicted in TABLE I, the rotor position parameter and the x-, y-axis coordinates. It should be noted that the neural network does not initially possess the knowledge that 0 and 2π are adjacent, and the training would struggle to learn this without preprocessing. Therefore, the rotor position θ_e is transformed into the sin(θ_e) and cos(θ_e) function. For the x-axis and y-axis coordinates processing, two model structures are comparatively proposed as shown in Fig. 3. Model in Fig. 3(a) utilizes the direct input of the x-axis and y-axis coordinates, whilst the model in Fig. 3(b) adds one constant Fourier layer that transformed x-axis and y-axis coordinates into 128 pairs of the sin and cos function values. Various frequency components are included in the Fourier layer, and therefore



(a) Deep learning based surrogate model structure without Fourier layer



(b) Deep learning based surrogate model structure with Fourier layer

Fig. 3. Illustration of the deep learning based surrogate model structure.

both low frequency and high frequency physics can be reflected in the model with Fourier layer processing.

It should be noted that different from the physics informed neural network, which explicitly embed the knowledge of the boundary and the excitation conditions in the neural network. The proposed model utilizes the implicit way for embedding the neural network, which will directly learn the boundary conditions and the excitation conditions through the physics pattern embedded in the dataset.

The training process is established based on the calculation platform with NVIDIA RTX 5090, AMD EPYC 7543 32-core CPU and 117 GB RAM. It should be noted that the training environment can be different from the inference environment, and the proposed model performance does not require state of art graphical processing unit (GPU) during the model inference stage. The training time for 100 epoch will consume about 7.78 hours. Once obtained the trained deep learning surrogate model, there is no need to retrain the model at the inference state.

The training loss and the validation loss for the two proposed deep learning surrogate models are shown in Fig. 4. It shows that the embedded Fourier layer could efficiently improve both the training loss and the validation loss of the deep learning surrogate model. The training would last for 100 epochs, and the Fourier layer embedded surrogate model is more stable than that with direct x-, y-axis coordinates input one. Since the neural network always tends to fit the target through a smooth function. However, the magnetic field distribution has sharp and clear edge due to the existence of material interface. It will make the conventional deep learning model struggle to find the optimal parameters.

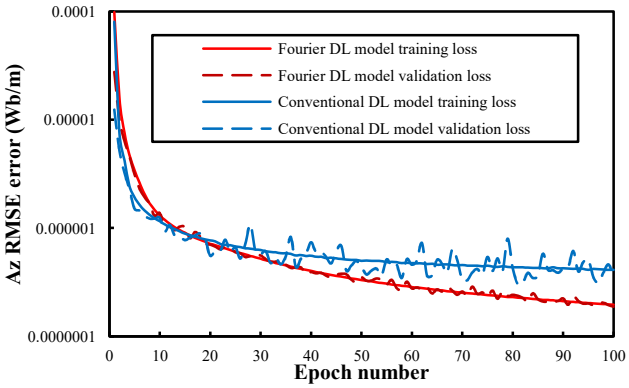


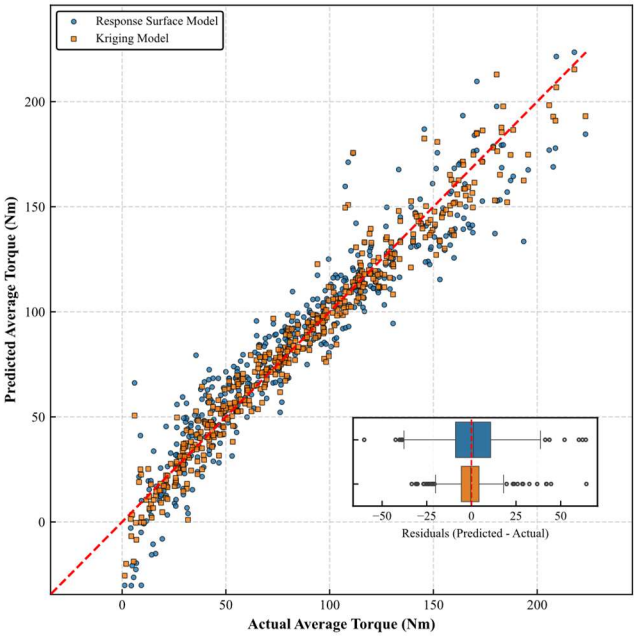
Fig. 4. Training loss and validation loss for the two proposed models.

C. Comparison of Surrogate Models

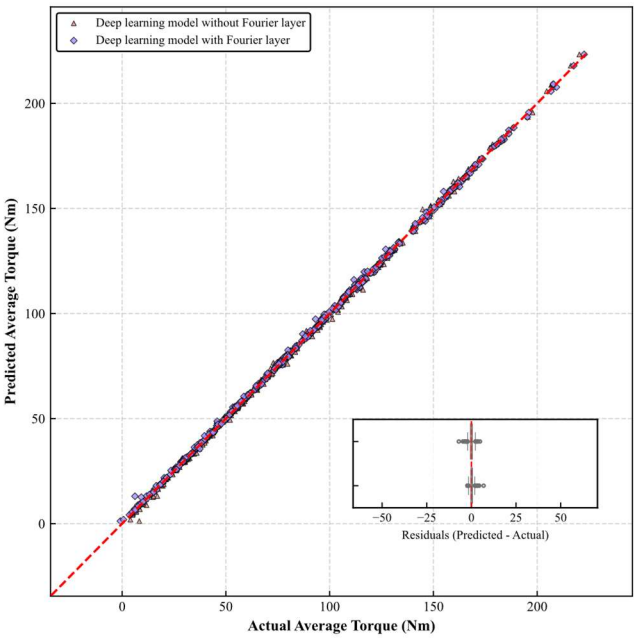
In this sub-section, the average torque and the iron loss prediction capability will be evaluated in both conventional surrogate models and the deep learning surrogate models. The conventional surrogate model includes the response surface model and the Kriging model.

It should be noted that the deep learning surrogate models only predict the magnetic vector potential distribution, and the average torque and the iron loss are evaluated based on the post-processing method in the finite element analysis method.

Fig. 5 shows the comparison of the average torque prediction accuracy for all the surrogate models. It is evident that the deep learning based surrogate models have a higher accuracy than the conventional surrogate models. It can be reflected through the box plot of the torque prediction residuals. In the conventional surrogate models, the Kriging model has a better prediction accuracy than the response surface model. As for the deep learning based models, both surrogate models have similar torque prediction accuracy.

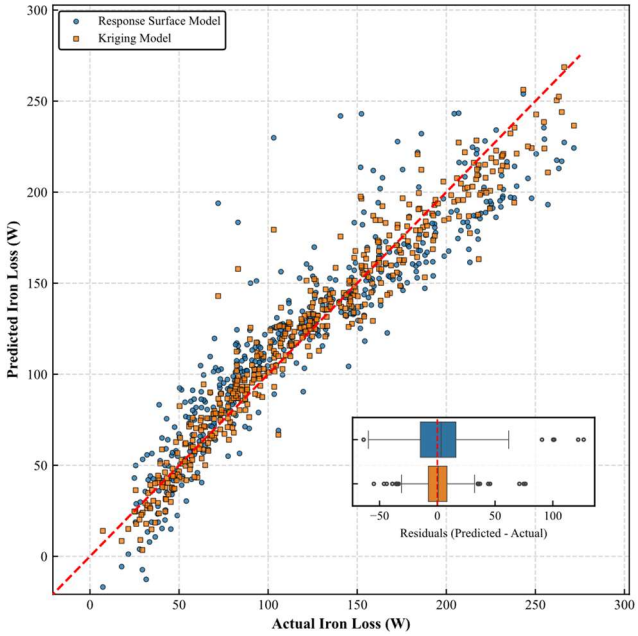


(a) Conventional surrogate models

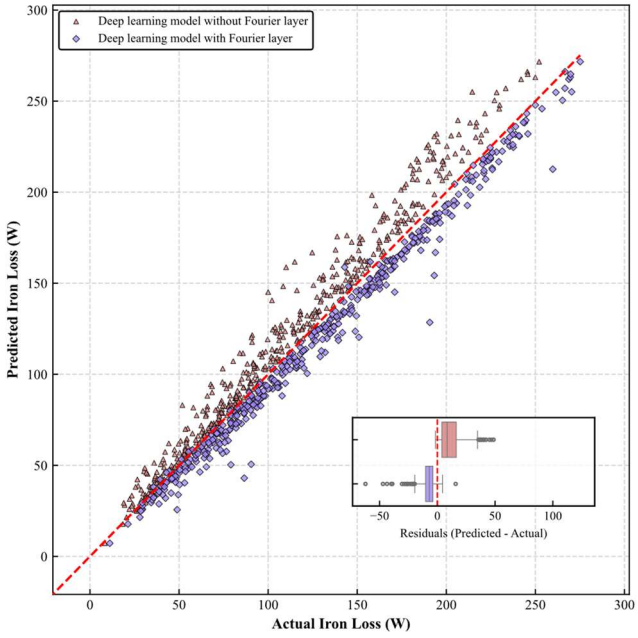


(b) Deep learning surrogate models

Fig. 5. Torque prediction accuracy for conventional surrogate models.



(a) Conventional surrogate models



(b) Deep learning surrogate models

Fig. 6. Iron loss prediction accuracy for conventional surrogate models.

As for the iron loss prediction accuracy, the deep learning based surrogate models also have a better performance than the Kriging and response surface model. The deep learning surrogate model with Fourier layer embedded has a better prediction accuracy than that without Fourier layer.

It should be noted that the deep learning based surrogate model has a slight loss prediction offset in both models. Since the iron loss is sensitive to the local magnetic field distribution, the iron loss prediction difficulty is higher than that in the torque prediction. The unclear edge flux density or local fluctuating flux density would have made iron loss prediction error, and the detailed analysis will be given in the magnetic field prediction fidelity analysis in the next sub-section.

The R^2 value of all the surrogate models are shown in TABLE II, it shows the deep learning model has a superior performance in the average torque prediction. As for the iron loss prediction, the kriging model also has a high prediction accuracy in the conventional models, and the deep learning model with Fourier layer has the best accuracy among 4 models.

TABLE II R^2 VALUE OF THE SURROGATE MODELS

Model	Average torque	Iron loss
Response surface	0.8897	0.8332
Kriging	0.9533	0.9410
Deep learning without Fourier layer	0.9994	0.9297
Deep learning with Fourier layer	0.9996	0.9730

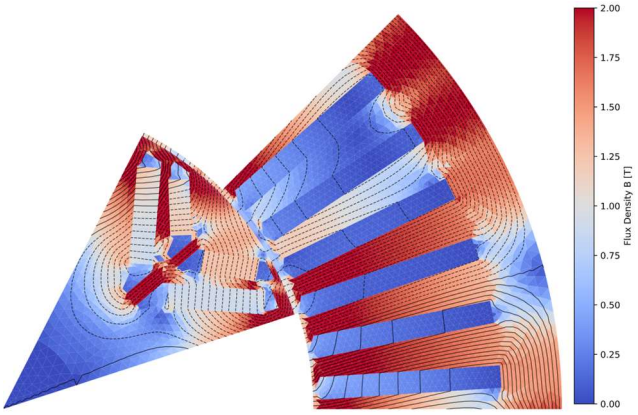
D. Fidelity Analysis

The proposed surrogate model can predict the magnetic field of the IPM machine, which is a unique characteristic compared to the traditional models. In this sub-section, the fidelity analysis will be given for both deep learning based surrogate models, including the magnetic field analysis and the iron loss distribution analysis.

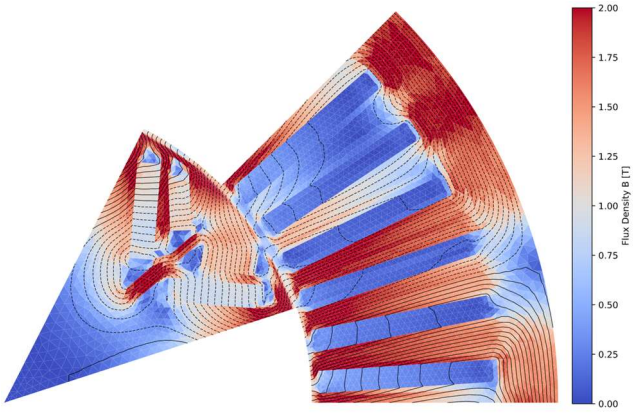
Choose one design case in the validation set for the verification, the magnetic field comparison is shown in Fig. 7 (the design case is not trained for the deep learning model). The proposed deep learning surrogate model has good agreement with the FE analysis. The magnetic field details are well captured including the material boundaries and the local magnetic saturation. However, it should be noted that the deep learning methods have a nonperfect magnetic flux density distribution in the stator slot and the rotor PM flux barriers. It is caused by the small fluctuation of the magnetic vector potential prediction in the neural networks.

Although not explicitly given to the deep learning model, the odd symmetry condition has been established in the proposed method. It proves that the implicit physics knowledge embedment is effective for deep learning surrogate models.

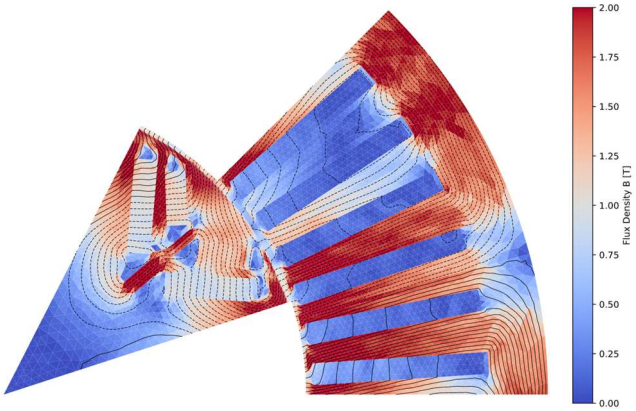
The iron loss prediction, which is calculated based on frequency Bertotti model considering the rotational loss correction, is shown in Fig. 8 including both FE analysis and surrogate models. It can be observed in FE analysis that the iron loss is significant in slot corner. However, the surrogate model is difficult to capture this detail. Moreover, the iron loss distribution in the stator slot suffers from poor uniform characteristic for the deep learning based surrogate models.



(a) FE analysis

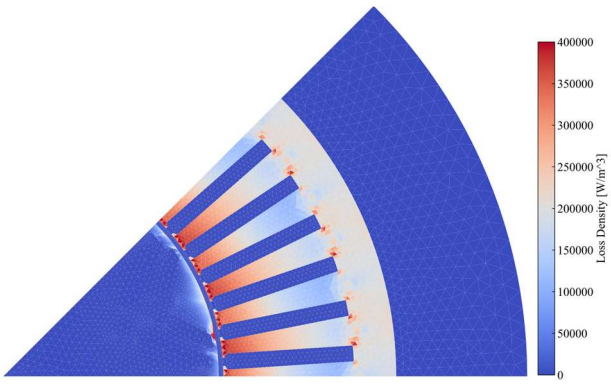


(b) Deep learning surrogate models without Fourier layer

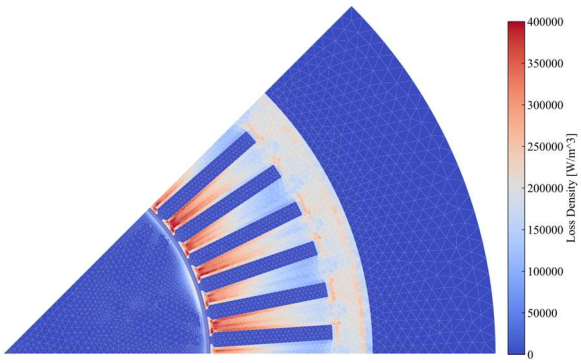


(c) Deep learning surrogate models with Fourier layer

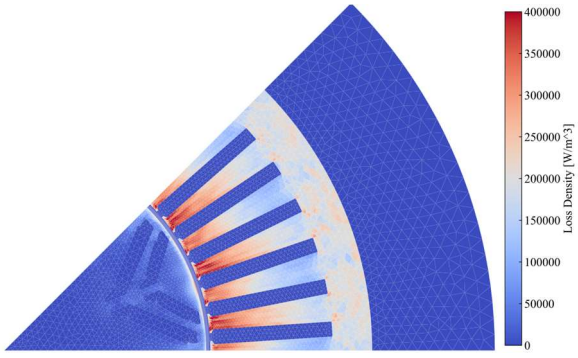
Fig. 7. Magnetic field distributions of surrogate models and FE analysis.



(a) FE analysis



(b) Deep learning surrogate models without Fourier layer



(c) Deep learning surrogate models with Fourier layer

Fig. 8. Iron loss distributions of surrogate models and FE analysis.

E. Calculation Time Comparison

The deep learning model reference time is compared with the FE calculation time in Fig. 9 (~10000 mesh nodes). It shows that the deep learning model inference time for 60 time steps in IPM case analysis consumes 0.578s, whilst that in FE analysis consumes 11.1 s. The deep learning based surrogate model has much lower calculation time than the FE analysis, which can be an attractive option for the alternative analysis tool for the EV used traction machine.

It should be noted that the model inference is implemented on low-cost NVIDIA RTX 2070 Super. The training state and the inference state of the deep learning surrogate model can be established on different GPU hardware, and the NVIDIA RTX 2070 Super is much affordable for the RTX 5090.

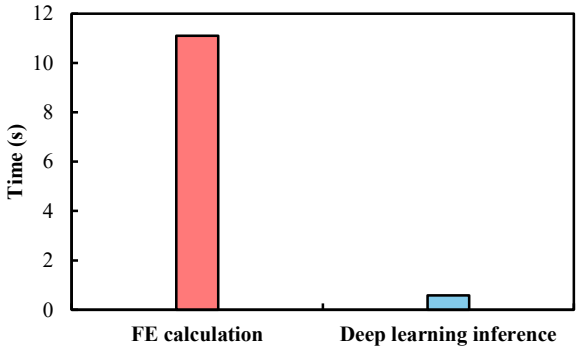


Fig. 9. Calculation time comparison for 60 time steps IPM case analysis.
* Deep learning model is inferred based on NVIDIA RTX 2070 Super
** FE calculation is based on Intel Core Ultra 9 285H

IV. Conclusion and Discussions

In this paper, a deep learning based surrogate modeling method is proposed for the EV used IPM traction machine. Different from the traditional surrogate models that predicts the performance of the machine based on the design parameters, the proposed method will predict the magnetic vector potential distribution of the IPM machine under various rotor positions. The prediction results shows that the proposed deep learning based surrogate model has superior performance in the average torque and iron loss prediction than the traditional surrogate models. As for the calculation time, it is >94% reduced compared with the FE calculation. All the analysis and performance comparison are verified in the validation set, which is not utilized in the neural network training.

This paper has proved that the deep learning analysis approach can be a powerful tool for the magnetic field analysis and prediction for the EV used traction machine. The related analysis and modeling methodologies can guide future design and optimization patterns based on the advanced data driven modeling approach. It should be noted that the input dimension of the analysed IPM machine is 15, which features a very high dimensional surrogate modeling. Considering the output of the surrogate model is the magnetic vector potential with high spatial resolution, it is impossible for the deep learning network to overfitting every details of the magnetic field. Therefore, the finally obtained network model has high generalization capability, which can be reflected through the stable decrease of the RMSE error in both training dataset and the validation set. Future works can focus on extending the modeling approach for wider application scenarios and better prediction accuracy.

Appendix

The magnetostatic govern equation in the analysed IPM machine can be expressed as,

$$\nabla \times (\upsilon \nabla \times A) = J + \nabla \times (\upsilon B_r) \quad (1)$$

where A is the magnetic vector potential, J is the source current density, B_r is the remanent magnetic flux density vector of the magnets, υ is the magnetic reluctivity of the material. In 2D electrical machine analysis, both J and A have only the z-axis components.

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BIOGRAPHIES

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